

Pranav Nagarajan

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Education

California Institute of Technology	Pasadena, CA
<i>Ph.D. in Astrophysics</i>	<i>Expected 2027</i>
<i>M.S. in Astrophysics</i>	<i>2024</i>
Advisor: Kareem El-Badry	GPA: 4.2/4.0
University of California, Berkeley	Berkeley, CA
<i>B.A. in Astrophysics, Physics, and Data Science</i>	<i>2022</i>
Highest Distinction in General Scholarship	GPA: 4.0/4.0

Research Experience

Graduate Student Researcher	California Institute of Technology
<i>Advisor: Prof. Kareem El-Badry</i>	<i>2022-Present</i>
Discovering and Characterizing Stellar-Mass Black Holes and Neutron Stars in Galactic Binaries	
Undergraduate Student Researcher	University of California, Berkeley
<i>Advisor: Prof. Daniel Weisz</i>	<i>2020-2022</i>
Mapping the Local Group with RR Lyrae and Bayesian Hierarchical Modeling	
BSRC Summer Internship	Berkeley SETI Research Center
<i>Mentor: Dr. Vishal Gajjar</i>	<i>Summer 2021</i>
Breakthrough Listen Periodic Spectral Signals Search	
Undergraduate Research Apprenticeship Program	Space Sciences Laboratory
<i>Mentor: Dr. Andreas Zoglauer</i>	<i>2019-2021</i>
Using 3D Convolutional Neural Networks and Graph Neural Networks for Compton Track Identification in Gamma-Ray Space Telescopes	

Technical Skills

- ❖ Coding: Python (pandas, Matplotlib, NumPy, SciPy, scikit-learn, Keras, TensorFlow), SQL (ADQL), Jupyter Notebook, Git, L^AT_EX
- ❖ Statistical Modeling: Bayesian Inference, Markov Chain Monte Carlo, Hypothesis Testing, Time Series Analysis, Machine Learning

Teaching Experience

Teaching Assistant

High-Energy Astrophysics, Galaxies and Cosmology

California Institute of Technology

2024

Undergraduate Student Instructor

Principles and Techniques of Data Science

University of California, Berkeley

2020

Honors & Awards

Neugebauer Scholar	2025
NSF Graduate Research Fellowship Program: Honorable Mention	2024
URAP Summer Award	2020
National Merit Scholarship	2018

Awarded Telescope Time

Palomar Hale 200-inch	5 nights (PI) + 3 nights (Co-I)
Keck 10m	9 nights (Co-I)

Observing Experience

Keck I Telescope, High Resolution Echelle Spectrometer – 1 night	2025
Keck II Telescope, Near-Infrared Camera 2 – 3.5 nights	2024–2025
Keck II Telescope, Near-Infrared Echellete Spectrometer – 3 nights	2024
Very Large Telescope, Focal Reducer and Low Dispersion Spectrograph 2 – 1 night	2024
Magellan I Telescope, Inamori Magellan Areal Camera and Spectrograph – 1 night	2024
Palomar Hale Telescope, Wide-Field Infrared Camera – 2 nights	2024
Keck I Telescope, Low Resolution Imaging Spectrometer – 1 night	2024
Palomar Hale Telescope, Caltech High-speed Multi-color Camera – 1 night	2023
Palomar Hale Telescope, Double Spectrograph – 3.5 nights	2023
Keck II Telescope, Echellete Spectrograph and Imager – 1 night	2022

Journal Referee

MNRAS, ApJ, AJ, OJAp, ApJL, New Astron., Sci. Bull.	8 papers total, 2023 – Present
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Talks

Talk, Keck Science Meeting 2025, Los Angeles, CA	Sep. 2025
Talk, Binary Stars in the Space Era conference, Keele, United Kingdom	Jul. 2025
Talk, Symbiotic stars conference, Prague, Czech Republic	Jun. 2024
Tea Talk, Observatories of the Carnegie Institution for Science, Pasadena, CA	May 2024
Talk, Stellar-Mass Black Holes Group Meeting, Pasadena, CA	Jan. 2024
Talk, Zwicky Transient Facility Team Meeting, Pasadena, CA	Oct. 2023
Talk, Zwicky Transient Facility Stellar Group Meeting, Pasadena, CA	Mar. 2023

Refereed Publications

13. **Nagarajan, P.**, El-Badry, K., Reggiani, H., Lam, C. Y., Simon, J. D., Müller-Horn, J., Seeburger, R., Rix, H.-W., Isaacson, H., Lu, J. R., Chandra, V., and Andrae, R. (2025). A Spectroscopic Search for Dormant Black Holes in Low-Metallicity Binaries. *Publications of the Astronomical Society of the Pacific*, 137(9), 094202.
12. Galiullin, I., Rodriguez, A. C., El-Badry, K., Caiazzo, I., Szkody, P., **Nagarajan, P.**, and Whitebook, S. (2025). Optical Spectroscopy of the Most Compact Accreting Binary Harboring a Magnetic White Dwarf and a Hydrogen-rich Donor. *The Astrophysical Journal Letters*, 990(2), L57.
11. **Nagarajan, P.**, El-Badry, K., Chawla, C., Di Carlo, N., Breivik, K., Rodriguez, C. L., Agrawal, P., Delfavero, V., & Chatterjee, S. (2025). Realistic Predictions for Gaia Black Hole Discoveries: Comparison of Isolated Binary and Dynamical Formation Models. *Publications of the Astronomical Society of the Pacific*, 137(4), 044202.
10. Hinkle, K., **Nagarajan, P.**, Fekel, F. C., Mikołajewska, J., Straniero, O., & Muterspaugh, M. W. (2025). Binary Parameters for the Recurrent Nova T Coronae Borealis. *The Astrophysical Journal*, 983(1), 76.
9. **Nagarajan, P.** & El-Badry, K. (2025). Mixed Origins: Strong Natal Kicks for Some Black Holes and None for Others. *Publications of the Astronomical Society of the Pacific*, 137(3), 034203.
8. Blomberg, L., El-Badry K., Breivik, K., Caiazzo, I., **Nagarajan, P.**, Rodriguez, A., van Roestel, J., Vanderbosch, Z. P., & Yamaguchi, N. (2024). The Companion Mass Distribution of Post Common Envelope Hot Subdwarf Binaries: Evidence for Boosted and Disrupted Magnetic Braking? *Publications of the Astronomical Society of the Pacific*, 136(12), 124201.
7. **Nagarajan, P.** & El-Badry, K. (2024). Validation of Gaia DR3 orbital and acceleration solutions with hierarchical triples. *Publications of the Astronomical Society of the Pacific*, 136(9), 094203.
6. **Nagarajan, P.**, El-Badry, K., Lam, Y. C., & Reggiani, H. (2024). The Symbiotic X-ray Binary IGR J16194-2810: A Window on the Future Evolution of Wide Neutron Star Binaries From Gaia. *Publications of the Astronomical Society of the Pacific*, 136(7), 074202.
5. **Nagarajan, P.**, El-Badry, K., Triaud, A. H., Baycroft, T. A., Latham, D., Bieryla, A., Buchhave, L. A., Rix, H.-W., Quataert, E., Howard, A., Isaacson, H., & Hobson, M. J. (2024). ESPRESSO Observations of Gaia BH1: High-precision Orbital Constraints and no Evidence for an Inner Binary. *Publications of the Astronomical Society of the Pacific*, 136(1), 014202.
4. **Nagarajan, P.**, El-Badry, K., Rodriguez, A. C., van Roestel, J., & Roulston, B. (2023). Spectroscopic follow-up of black hole and neutron star candidates in ellipsoidal variables from *Gaia* DR3. *Monthly Notices of the Royal Astronomical Society*, 524(3), 4367–4383.
3. El-Badry, K., Shen, K. J., Chandra, V., Bauer, E. B., Fuller, J., Strader, J., Chomiuk, L., Naidu, R. P., Caiazzo, I., Rodriguez, A. C., **Nagarajan, P.**, Yamaguchi, N., Vanderbosch, Z. P., Roulston, B. R., Gänsicke, B., Han, J. J., Burdge, K. B., Filippenko, A. V., Brink, T. G., & Zheng, W. (2023). The fastest stars in the galaxy. *The Open Journal of Astrophysics*, 6.
2. Suresh, A., Gajjar, V., **Nagarajan, P.**, Sheikh, S. Z., Siemion, A. P., Lebofsky, M., MacMahon, D. H., Price, D. C., & Croft, S. (2023). A 4–8 GHz Galactic Center Search for Periodic Technosignatures. *The Astronomical Journal*, 165(6), 255.

1. **Nagarajan, P.**, Weisz, D., & El-Badry, K. (2022). RR Lyrae-based distances for 39 nearby dwarf galaxies calibrated to *Gaia* EDR3. *The Astrophysical Journal*, *932*(1), 19.